

- 5 During the operation of a refrigerator, the coolant is constantly being liquefied under pressure, and vaporised once the pressure is released. Under these conditions the gas does not behave as an ideal gas.

Which statement best describes why the gas liquefies under pressure?

- A Increasing the pressure causes the temperature of the gas to decrease to below its boiling point.
- B Increasing the pressure decreases the kinetic energy of the molecules, creating the ordered liquid state.
- C Increasing the pressure decreases the size of the molecules, and so increases the intermolecular attractions.
- D Increasing the pressure pushes the molecules closer together, and so the short-range intermolecular forces of attraction are more effective.